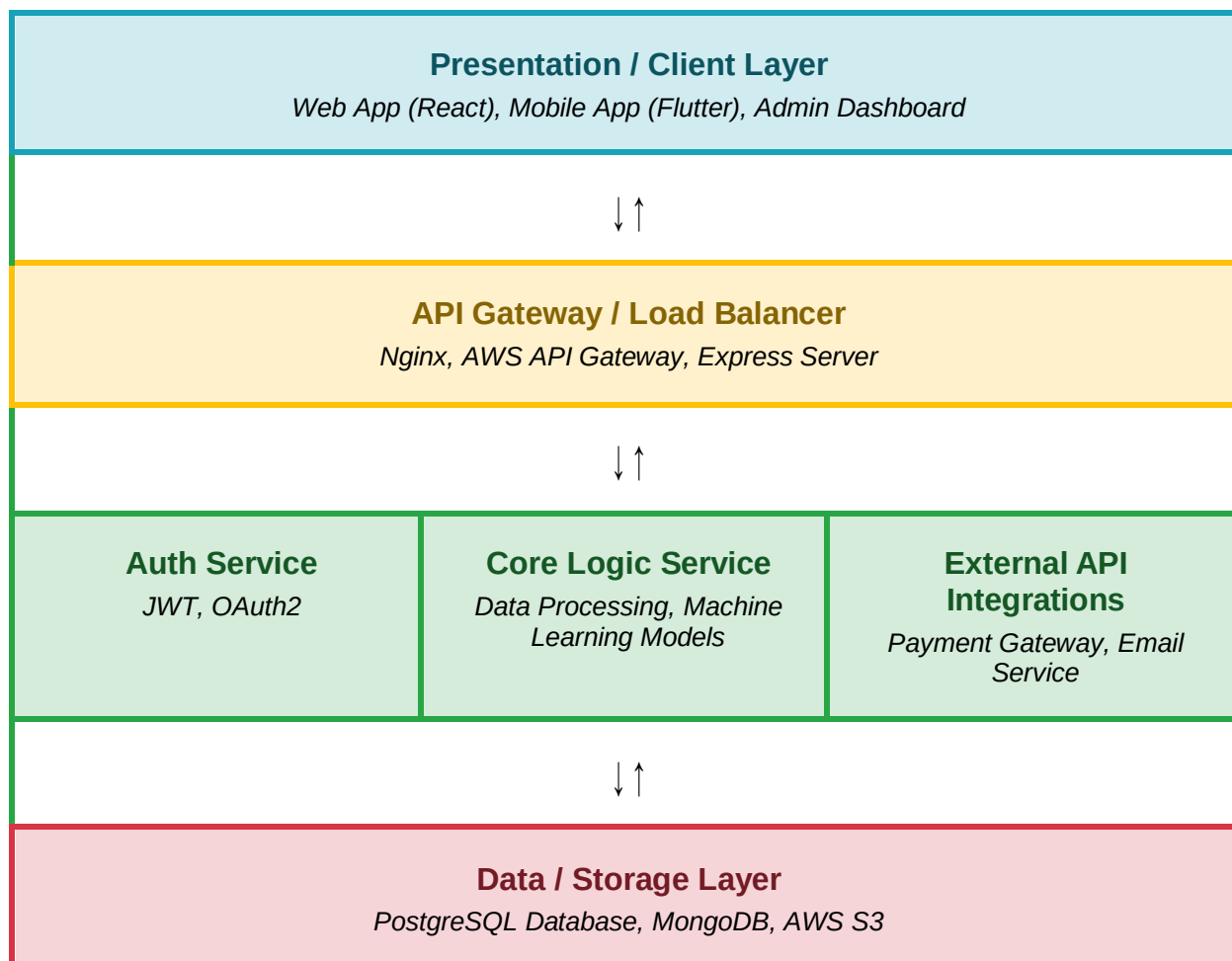


Solution Architecture

Date	05 JULY 2026
Team ID	SWTID-2026-2423
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	<i>Serves as the user interface where loan applicants input their details (e.g., income, credit history) and view their real-time loan approval prediction status.</i>	HTML5, CSS3, JavaScript (or standard Jinja2 templates integrated with Flask)
API Gateway	<i>Acts as the entry point that routes client requests from the presentation layer to the backend application logic safely and handles request traffic.</i>	Flask Built-in Routing / Nginx (as a reverse proxy if deployed)
Microservices / Core Logic	<i>Processes incoming applicant data, executes data preprocessing, and runs the trained machine learning model to evaluate creditworthiness and make real-time approval decisions.</i>	Python, Flask, scikit-learn, pandas, numpy
Database / Storage Layer	<i>Stores historical applicant records, training datasets, system logs, and serialized machine learning model files (.pkl) for continuous learning.</i>	Local/Cloud Storage (1 TB SSD) for handling Kaggle and UCI CSV datasets